

templates\learn\cve-vs-cwe-vs-cvss.html

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{% extends "base.html" %}
{% block content %}

<!-- Embedded CSS styles for this specific educational page -->
<style>
/* Container for Venn diagram and legend layout */
.venn-container {
display: flex;
align-items: center;
justify-content: center;
margin: 22px 0;
gap: 40px; /* Space between diagram and legend */
}

/* Main Venn diagram container with relative positioning */
.venn-diagram {
position: relative;
width: 450px;
height: 350px;
padding: 20px;
}

/* Base circle styling for all three frameworks */
.circle {
position: absolute; /* Allows precise positioning for overlap */
border-radius: 50%; /* Creates perfect circles */
opacity: 0.85; /* Transparency for intersection visibility */
display: flex;
align-items: center;
justify-content: center;
font-weight: bold;
font-size: 12px;
text-align: center;
line-height: 1.2;
border: 3px solid; /* Border color set individually */
}

/* CVE circle - Blue theme for vulnerability identifiers */
.cve-circle {
width: 180px;
height: 180px;
background: rgba(99, 164, 255, 0.6); /* Semi-transparent blue */
border-color: #63a4ff;
top: 30px;
left: 60px; /* Positioned to overlap with other circles */
color: #63a4ff;
}

/* CWE circle - Teal theme for weakness categories */
.cwe-circle {
width: 180px;
height: 180px;
background: rgba(56, 178, 172, 0.6); /* Semi-transparent teal */
border-color: #38b2ac;
top: 30px;
left: 190px; /* Overlaps with CVE circle */
color: #38b2ac;
}

/* CVSS circle - Red theme for severity scoring */
.cvss-circle {
width: 180px;
height: 180px;
background: rgba(245, 101, 101, 0.6); /* Semi-transparent red */
border-color: #f56565;
top: 150px;
left: 125px; /* Positioned to create triple intersection */
color: #f56565;
}

/* Framework name labels positioned on circles */
.label {
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position: absolute;
font-weight: bold;
font-size: 14px;
z-index: 10; /* Appear above circles */
}

/* Individual label positioning for each framework */
.cve-label {
top: 70px;
left: 105px;
color: #63a4ff;
}

.cwe-label {
top: 70px;
left: 255px;
color: #38b2ac;
}

.cvss-label {
top: 250px;
left: 165px;
color: #f56565;
}

/* Example identifiers for each framework */
.example {
position: absolute;
font-size: 10px;
background: rgba(0, 0, 0, 0.8); /* Dark background for readability */
padding: 3px 6px;
border-radius: 4px;
white-space: nowrap; /* Prevent text wrapping */
z-index: 5;
}

/* Example positioning for each framework */
.cve-example {
top: 95px;
left: 75px;
color: #e2e8f0;
}

.cwe-example {
top: 95px;
left: 270px;
color: #e2e8f0;
}

.cvss-example {
top: 275px;
left: 150px;
color: #e2e8f0;
}

/* Labels for intersection areas showing relationships */
.intersection-label {
position: absolute;
font-size: 10px;
background: rgba(255, 255, 255, 0.9); /* High contrast for visibility */
color: #1a202c;
padding: 3px 6px;
border-radius: 4px;
font-weight: 600;
z-index: 15; /* Above all other elements */
}

/* Positioning for two-way intersections */
.cve-cwe {
top: 115px;
left: 185px;
}

.cve-cvss {
top: 175px;
left: 115px;
}

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.cwe-cvss {
top: 175px;
left: 255px;
}

/* Center intersection where all three frameworks meet */
.center-intersection {
top: 165px;
left: 170px;
background: rgba(255, 215, 0, 0.9); /* Gold for emphasis */
color: #1a202c;
font-size: 9px;
}

/* Legend styling with cybersecurity theme */
.legend {
color: #e3e8f0;
background: linear-gradient(135deg, #232b45 0%, #1e2640 100%); /* Dark gradient */
border-radius: 12px;
box-shadow: 0 6px 24px rgba(0,0,0,0.3); /* Subtle shadow */
border: 1px solid rgba(99, 164, 255, 0.1);
padding: 18px 24px;
margin: 0 0 18px 0;
width: 350px;
}

/* Legend title styling */
.legend-title {
font-weight: bold;
font-size: 16px;
margin-bottom: 14px;
color: #cdd7ff;
}

/* Individual legend items layout */
.legend-item {
display: flex;
align-items: center;
margin-bottom: 8px;
}

/* Small colored circles in legend */
.legend-circle {
width: 16px;
height: 16px;
border-radius: 50%;
margin-right: 12px;
border: 2px solid #333;
flex-shrink: 0; /* Maintain size */
}

/* Legend circle colors matching main diagram */
.legend-cve {
background: rgba(99, 164, 255, 0.6);
border-color: #63a4ff;
}

.legend-cwe {
background: rgba(56, 178, 172, 0.6);
border-color: #38b2ac;
}

.legend-cvss {
background: rgba(245, 101, 101, 0.6);
border-color: #f56565;
}

/* Legend text styling */
.legend-text {
font-size: 13px;
}

.legend-name {
margin-bottom: 0;
}

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/* Warning items get red color treatment */
.warning-item .legend-name {
color: #e74c3c;
}
</style>

<!-- Main content container with max width and centering -->
<div style="max-width:1400px;margin:0 auto 32px auto;padding-top:18px;">
<!-- Page title with CVE blue color scheme -->
<h1 style="color:#F5E8D8;">CVE vs CWE vs CVSS</h1>

<!-- Introduction paragraph explaining the three frameworks -->
<p style="color:#e2e8f0;font-size:1.07em;">
Vulnerability analysis uses three frame-works: <b>CVE</b>, <b>CWE</b>, and <b>CVSS</b>
each provides a unique way to understand risk and weaknesses.
</p>

<!-- Comparison table showing framework roles and examples -->
<table style="color:#e3e8f0;background:linear-gradient(135deg, #232b45 0%, #1e2640
100%);border-radius:12px;width:75%;margin-bottom:18px; border-collapse:collapse;border:
1px solid rgba(99, 164, 255, 0.1);box-shadow: 0 6px 24px rgba(0,0,0,0.3);">
<!-- Table header with framework categories -->
<tr style="background:#232b45;">
<th style="text-align:center;padding:12px 8px;border-radius:12px 0 0 0;">Framework</th>
<th style="text-align:center;padding:12px 8px;">Role in Security</th>
<th style="text-align:center;padding:12px 8px;border-radius:0 12px 0 0;">Sample</th>
</tr>
<!-- CVE row with blue color scheme -->
<tr style="background:rgba(38, 49, 89, 0.4);">
<td style="text-align:center;padding:10px 8px;"><strong style="color:
#fcfcfc;">CVE</strong></td>
<td style="text-align:center;padding:10px 8px;">Specific vulnerability identifier</td>
<td style="text-align:center;padding:10px 8px;">CVE-2021-34527</td>
</tr>
<!-- CWE row with teal color scheme -->
<tr style="background:rgba(38, 49, 89, 0.4);">
<td style="text-align:center;padding:10px 8px;"><strong
style="color:#fcfcfc;">CWE</strong></td>
<td style="text-align:center;padding:10px 8px;">Type/classification of weakness</td>
<td style="text-align:center;padding:10px 8px;">CWE-79 (Cross-Site Scripting)</td>
</tr>
<!-- CVSS row with red color scheme -->
<tr style="background:rgba(38, 49, 89, 0.4);">
<td style="text-align:center;padding:10px 8px;border-radius:0 0 0 12px;"><strong
style="color:#fcfcfc;">CVSS</strong></td>
<td style="text-align:center;padding:10px 8px;">Numerical severity score</td>
<td style="text-align:center;padding:10px 8px;border-radius:0 0 12px 0;">9.8
(Critical)</td>
</tr>
</table>

<!-- Section explaining framework relationships -->
<h2 style="color:#cdd7ff;margin-top:32px;">How Are They Connected?</h2>
<ul style="color:#e3e8f0;margin-left:17px;">
<li><b>CVE</b> records the bug.</li>
<li><b>CWE</b> classifies what kind of bug it is.</li>
<li><b>CVSS</b> tells you how dangerous it is.</li>
</ul>

<!-- Venn diagram section header -->
<h2 style="color:#cdd7ff;">Visual Relationship (Venn Diagram)</h2>
<div class="venn-container">
<!-- Main Venn diagram with three overlapping circles -->
<div class="venn-diagram">
<!-- Three framework circles with transparency for overlaps -->
<div class="circle cve-circle"></div>
<div class="circle cwe-circle"></div>
<div class="circle cvss-circle"></div>

<!-- Framework name labels positioned on circles -->
<div class="label cve-label">CVE</div>
<div class="label cwe-label">CWE</div>
<div class="label cvss-label">CVSS</div>

<!-- Example identifiers for each framework -->
<div class="example cve-example">CVE-2021-34527</div>
<div class="example cwe-example">CWE-79</div>

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<div class="example cvss-example">9.8 Critical</div>

<!-- Labels showing what each intersection represents -->
<div class="intersection-label cve-cwe">ID + Type</div>
<div class="intersection-label cve-cvss">ID + Score</div>
<div class="intersection-label cwe-cvss">Type + Risk</div>
<div class="intersection-label center-intersection">Complete Analysis</div>
</div>

<!-- Legend explaining how to interpret the diagram -->
<div class="legend">
<div class="legend-title">How to read this diagram?</div>
<!-- General explanation of frameworks -->
<div class="legend-item cve-item">
<div class="legend-text">
<div class="legend-name">Each circle is a <strong style="color:
#3498db;">FRAMEWORK</strong>.</div>
</div>
</div>
<!-- CVE framework explanation with color coding -->
<div class="legend-item cve-item">
<div class="legend-circle legend-cve"></div>
<div class="legend-text">
<div class="legend-name"><strong style="color:#63a4ff;">CVE</strong> - Specific
vulnerability ID</div>
</div>
</div>
<!-- CWE framework explanation -->
<div class="legend-item cwe-item">
<div class="legend-circle legend-cwe"></div>
<div class="legend-text">
<div class="legend-name"><strong style="color:#38b2ac;">CWE</strong> - Weakness
category/type</div>
</div>
</div>
<!-- CVSS framework explanation -->
<div class="legend-item cvss-item">
<div class="legend-circle legend-cvss"></div>
<div class="legend-text">
<div class="legend-name"><strong style="color:#f56565;">CVSS</strong> - Severity score
(0-10)</div>
</div>
</div>
<!-- Explanation of intersection meanings -->
<div style="margin-top: 14px; font-size: 12px; color: #a9adc1;">
<strong>Overlaps show relationships:</strong><br>
? <strong>CVE + CWE:</strong> Specific vuln with its type<br>
? <strong>CVE + CVSS:</strong> Specific vuln with severity<br>
? <strong>CWE + CVSS:</strong> Weakness type and risk level<br>
? <strong>Center:</strong> Complete vulnerability analysis
</div>
</div>
</div>

<!-- Real-world example using Log4Shell vulnerability -->
<h2 style="color:#cdd7ff;margin-top:32px;">Real-World Example: Log4Shell</h2>
<div
style="background:#232b45;border-radius:12px;padding:24px;margin-bottom:24px;border-left
:4px solid #f47174;">
<!-- Three-column grid showing each framework's role -->
<div style="display:grid;grid-template-columns:1fr 1fr 1fr;gap:20px;color:#e3e8f0;">
<!-- CVE component breakdown -->
<div style="background:#263159;padding:18px;border-radius:8px;border-top:3px solid
#63a4ff;">
<h4 style="color:#d3bb7a;margin:0 0 10px 0;">CVE Component</h4>
<div
style="font-family:monospace;color:#cdd7ff;font-weight:bold;margin-bottom:8px;">CVE-2021-
44228</div>
<div style="font-size:0.95em;line-height:1.4;">
This is the unique identifier that security teams, researchers, and databases use to
track this specific vulnerability. When someone says "CVE-2021-44228," everyone knows
exactly which bug they're talking about.
</div>
</div>
<!-- CVE component breakdown -->
<div style="background:#263159;padding:18px;border-radius:8px;border-top:3px solid
#38b2ac;">

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<h4 style="color:#d3bb7a;margin:0 0 10px 0;">CWE Component</h4>
<div
style="font-family:monospace;color:#cdd7ff;font-weight:bold;margin-bottom:8px;">CWE-502:
Deserialization</div>
<div style="font-size:0.95em;line-height:1.4;">
This tells us the <em>type</em> of weakness. It's a deserialization flaw?meaning the
software unsafely processes external data. This helps developers understand the pattern
and prevent similar bugs.
</div>
</div>
<!-- CVSS component breakdown -->
<div style="background:#263159;padding:18px;border-radius:8px;border-top:3px solid
#f47174;">
<h4 style="color:#d3bb7a;margin:0 0 10px 0;">CVSS Component</h4>
<div
style="font-family:monospace;color:#cdd7ff;font-weight:bold;margin-bottom:8px;">CVSS
3.1: 10.0 (Critical)</div>
<div style="font-size:0.95em;line-height:1.4;">
This tells us <em>how dangerous</em> it is. A 10.0 score means maximum severity?easy to
exploit, severe impact. This helps prioritize patching efforts.
</div>
</div>
</div>
<!-- Summary showing how all frameworks work together -->
<div
style="margin-top:20px;padding:16px;background:#1a2236;border-radius:8px;color:#a9adc1;"
>
<strong style="color:#d2be9d;">Together:</strong> CVE-2021-44228 is a critical (CVSS
10.0) deserialization vulnerability (CWE-502) in Apache Log4j2 that allows remote code
execution. The combination tells the complete story: what it is, how bad it is, and what
type of fix is needed.
</div>
</div>

<!-- Navigation back to main dashboard -->
<div style="text-align:center; margin:2.5em 0 1em 0;">
<a href="{{ url_for('main.index') }}" style="color:#F5E8D8; padding:0.8em 2em;
border-radius:6px; font-weight:600; text-decoration:none;">
? Back to Dashboard
</a>
</div>
<!-- End of content block -->
{% endblock %}

```